Comparative Analysis of CNN And Random Forest for Fashion-MNIST Classification

Project: Computer Vision

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Github link: https://github.com/maryamrad99/Project-Computer-Vision-Fashion-MNIST

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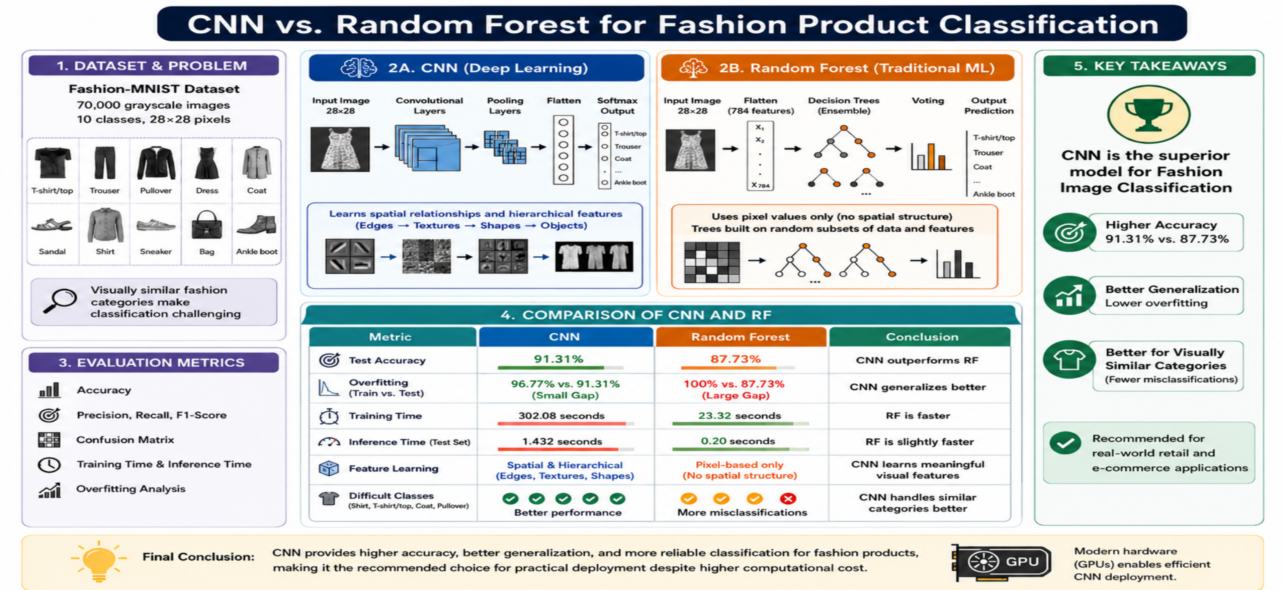
1. Abstract

1.1 Textual abstract

The experimental results demonstrate that the CNN achieved superior overall performance, with a training accuracy of 96.77% and a test accuracy of 91.31%, while the RF classifier achieved a training accuracy of 100% but a lower testing accuracy of approximately 87.73%, indicating overfitting. The confusion matrices, precision-recall, and F1-score analysis further reveal that both models struggled to classify visually similar upper-body garments; however, the CNN showcased significantly improved class separation by preserving spatial relationships. Although the RF classifier achieved faster training and inference times, the CNN provided higher predictive accuracy and more reliable generalization for fashion image classification applications.

1.2 Graphical Abstract

Fig.1 ***Comparison of CNN and Random Forest models for Fashion-MNIST classification***.



Source: Own finding. Created using Canva

2. Introduction & Literature review

2.1 Background

Artificial intelligence (AI) and machine learning (ML) technologies are widely used in the retail and e-commerce systems for tasks such as product recognition, inventory management, recommendation systems, and automated searches (Sridhar et al., 2026). One important application of AI in modern retail systems is image classification, where a computer system automatically is image classification, where computer systems automatically identify products from visual data.

This study uses the Fashion-MNIST dataset introduced by Xioa et al. (2017), which contains 70,000 greyscale fashion images across ten categories which include T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Bag, Sneaker, and Ankle boot. Each image has a resolution of 28x28 pixels and is divided into 60,000 training images and 10,000 testing images. The dataset is considered challenging because several clothing categories contain highly similar visual features, particulate Shirt, Pullover, Coat, and T-shirt/top (Xioa et al., 2017).

Convolutional Neural Networks (CNNs) are DL models designed for image processing that preserve spatial features through convolutional and pooling operations, allowing them to preserve spatial relationships within images (LeCun et al., 2015). In contrast, Random Forest (RF) is a traditional ML algorithm based on ensemble decision trees (Breiman, 2001). RF classifiers process images as flattened numerical vectors, which may reduce classification performance because spatial image information is lost during preprocessing.

2.2 Problem Statement

Fashion image classification remains challenging in computer vision (CV) because many clothing categories share overlapping visual patterns and structures, especially in low-resolution greyscale images (Xiao et al., 2017). Traditional ML algorithms such as Random Forest (RF) may struggle with these similarities because flattening images removes spatial relationships between pixels (Breiman, 2001). CNNs address this limitation by preserving spatial information and automatically extracting image features (LeCun et al., 2015).

This study aims to evaluate and compare the performance of a CNN and RF classifier using the Fashion-MNIST dataset developed by Xiao et al. (2017). The comparison focuses on the accuracy, precision, recall, F-1 score, confusion matrices, training performance, and generalization capability to determine which method is more suitable for automated fashion product recognition.

2.3 Related Research

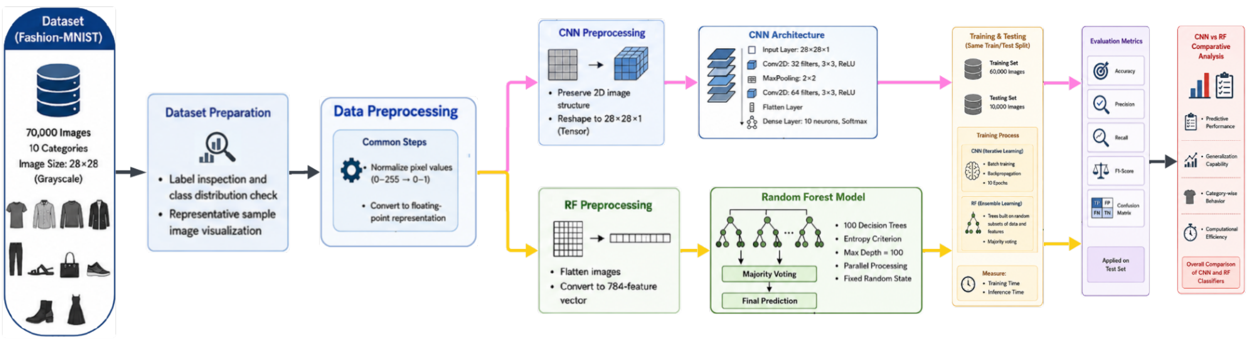
Previous research has shown that CNNs generally outperform traditional ML algorithms in image classification tasks. LeCun et al. (2015) and Goodfellow et al. (2017) highlighted the effectiveness of CNNs in learning spatial hierarchies and visual features from image data. Studies comparing CNNs with RF and other traditional methods reported that traditional models often struggle with visually similar image categories due to flattening images removing spatial information (Wang et al., 2021; Royan et al., 2025).

Research on Fashion-MNIST dataset also showcases that categories such as Shirt, Pullover, Coat, and T-shirt/top are frequently misclassified due to overlapping visual structures (Perera et al., 2025; Kayed et al., 2025).

3. Methodology

3.1 Graphical Methodology

Fig.2 ***Illustrates the overall workflow of the proposed CNN and RF classification methodology. The figure provides a comparative pipeline showing data preprocessing, model training, and evaluation stages for both approaches*.**



Source: Own finding, Created using Canva and Lucid.app

3.2 Dataset Preparation

The study was conducted using the Fashion-MNIST dataset introduced by Xiao et al. (2017), containing 70,000 greyscale images of fashion images divided into ten categories: T-shirt/top, Shirt, Pullover, Coat, Trouser, Sandal, Bag, Sneaker, Ankle boot, and Dress. Each image has a resolution of 28x28 pixels with intensity greyscale values ranging from 0 to 255.

The dataset was accessed via the TensorFlow (2024) implementation, using the predefined split of 60,000 training and 10,000 testing images. The dataset is balanced across all classes, ensuring equal representation and reducing classification bias (Xiao et al., 2017). Class labels were mapped to descriptive names for improved interpretability during visualization and evaluation.

3.3 Data Preprocessing

Data preprocessing was performed to improve the models performance, convergence, and its training efficiency. Since the CNN and RF classifier process image data differently, separate preprocessing procedures were conducted for each model.

All pixel values were normalized to a range of by dividing each pixel by 255, improving numerical stability and accelerating the models convergence during training (Salimans & Kingma, 2016).

For the CNN model, greyscale images were reshaped into 4D tensors with a single-image shape of 28x28x1 to preserve spatial relationships between the neighboring pixels, which is essential for convolutional operations. This allows the model to learn visual patterns such as edges, textures, and shapes through hierarchical feature extraction. The RF classifier could not directly process multi-dimensional image structures. Instead, each image was flattened 784-feature vectors, converting each 28x28 image into one-dimensional input suitable for traditional machine learning. All data were converted into floating-point numerical representation for compatibility with the training algorithms.

No additional augmentation techniques, such as image rotation, flipping, or scaling were applied, ensuring both models were evaluated under identical and controlled conditions.

3.4 CNN's Architecture

The CNN model was implemented using the Keras API provided by TensorFlow (2024) for multi-class classification. CNN's architecture is partially suitable for image classification tasks because convolutional operations preserve spatial relationships between neighboring pixels and enable hierarchical feature learning.

The architecture begins with an input layer for 28x28x1 images, followed by the first convolutional layer with 32 filters (3x3 kernel) and the Rectified Linear Unit (ReLU) activation function. The ReLU activation introduces non-linearity into the network and improves computational efficiency during training. A max-pooling layer (2x2) reduces spatial dimensions while preserving key features.

A second convolutional layer with 64 filters (3x3 kernel) was used to capture more complex patterns. Following this, the extracted feature maps were transformed into a one-dimensional vector using the flattening layer before being passed to a dense output layer with 10 neurons and Softmax activation function for class probability prediction.

The model was compiled using the Adam optimizer and sparse categorical cross-entropy loss function. The Adam optimizer was selected due to previous studies demonstrating its efficiency and stability for DL image classification tasks (Bera & Shrivastava, 2020) . During training, the model minimized classification error over 10 epochs through backpropagation.

3.5 Random Forest Configuration

The Random Forest (RF) classifier was implemented using the Scikit-learn framework as a traditional baseline model.

Kulkarni et al. (2014) described RF as "An ensemble algorithm," that "generates multiple decision trees as base classifiers and applies majority voting to combine the outcomes." Each decision tree is trained on random subsets of training data and feature space, while the final classification result is determined through majority voting among all trees within the forest. This ensemble strategy reduces variance and improves robustness compared to individual decision trees (Breiman, 2001).

Since RF cannot directly process two-dimensional image structures, all inputs were flattened into one-dimensional vectors consisting of 784 numerical pixel values prior to training. The model used 100 decision trees with entropy as the splitting criterion and a maximum tree depth of 100 to prevent excessive growth while capturing complex patterns. Parallel processing was used to improve training efficiency, and a fixed random state during the model initialization ensured reproducibility.

Unlike CNNs, RF does not perform feature extraction and instead relies entirely on the flattened pixel intensity values, which may reduce their performance on visually similar categories.

3.6 Training Strategy

Both models were trained using the same dataset split to ensure a fair and consistent comparison between the classifiers. The training process focused on evaluating each model's ability to learn meaningful representations from the dataset and generalize effectively to unseen test samples.

The CNN was trained using the normalized two-dimensional image dataset for a fixed number of epochs. During each epoch, batches of training images were processed and network weights were updated using backpropagation and the Adam optimizer. Sparse categorical cross-entropy was used as the loss function because the task involved multiple categorical labels represented as integer values. Batch-based learning improved computational efficiency and enabled stable gradient optimization throughout training.

The RF classifier was trained using bootstrapped subsets of the dataset, where each decision tree independently learned classification rules from the flattened image vectors, while the final predictions were determined through majority voting across all the estimators.

Training time for both the classifiers were measured to compare computational performance in addition to the predictive accuracy. And to ensure consistency and reproducibility, the same training and test partitions were maintained throughout the whole experiment to allow a direct comparison of the learning capability, generalization performance, and computational characteristics of both the classification approaches.

3.7 Evaluation Metrics

The performance of the CNN and RF classifier were evaluated using multiple metrics to provide a comprehensive analysis of classification effectiveness.

Accuracy was used as the primary metric, measuring overall correct classified predictions. Both training accuracy and test accuracy were calculated to assess the model's learning behavior and generalization performance. Precision, recall and the F1-score metrics were computed for each fashion category to provide a deeper insight into the class-specific performance. Precision measures correctness of positive predictions for a particular class, while recall measures how many actual samples were correctly identified. The F1-score provides a balanced metric of both.

Confusion matrices were also used to visualize the classification patterns across all categories, particularly between visually similar fashion products.

Finally, training and inference time were measured to evaluate computational efficiency between and real-world applicability of both models.

3.8 Use of AI-Assisted Writing Tools

During the development of this report, ChatGPT and Gemini were used as a supporting tool to evaluate draft sections, improve clarity, and assist in summarizing content to ensure terseness and alignment with the required academic structure. The tools were used to refine wording and help organize sections more effectively without altering the technical meaning of the word. All experimental design, implementation, and analysis were independently developed and verified by the author.

4. Results & Analysis

4.1 Accuracy Results

The performance of the CNN and RF classifier were evaluated using training accuracy, test accuracy, training time, and inference time. Both models were trained on the same Fashion-MNIST dataset to ensure a fair comparison between the deep learning DL and traditional ML approaches.

The CNN model achieved a training accuracy of 96.77% and a testing accuracy of 91.31%, showcasing that the model successfully learns meaningful spatial representation from the image data while maintaining strong predictive performance on previously unseen samples. In comparison, the RF classifier achieved a perfect accuracy of 100%, indicating that the model successfully memorized the training dataset, but the testing accuracy decreased to 87.73%, revealing weaker generalization performance, suggesting that the RF classifier experienced overfitting during training.

CNN's superior performance is due to its ability to preserve spatial relationships between neighboring pixels through convolutional operations and learn hierarchical visual features, whereas RF uses flattened pixel vectors that remove spatial structure.

Table 1: ***Performance Comparison Between CNN and RF***

Model	Training Time (s)	Inference Time (s)	Training Accuracy	Test Accuracy
RF	23.3220	0.2038	1.0000	0.8773
CNN	302.0886	1.4324	0.9677	0.9131

Note: From "CV_MNIST_FASHION.ipynb" file, section 4. Source: finding of author

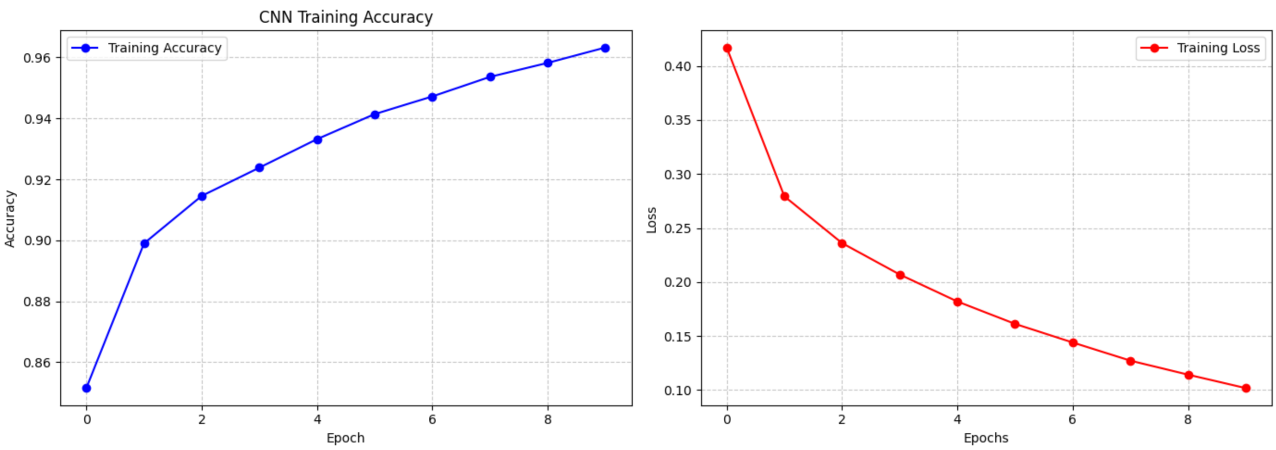
In terms of computational cost, the RF required significantly less training time than the CNN model. The RF completed training in approximately 23 seconds, whereas the CNN required around 302 seconds due to the convolutional operations and iterative optimization.

Although the RF classifier demonstrated faster computational performance, the CNN achieved a higher classification accuracy and better generalization capability. Therefore, the CNN provided a more reliable solution for fashion image classification despite its higher computational cost.

4.2 Training Curves

The learning behavior of the CNN was analyzed using the training accuracy and training loss curves recorded across ten training epochs. These curves provided insight into convergence, optimization stability, and overall learning performance.

Fig.3 CNN Training Accuracy and Loss Curves of CNN algorithm



Note: From "CV_MNIST_FASHION.ipynb" file, Section 2, Source: finding of author

The training accuracy curve shows a consistent upward trend, indicating improved classification performance over time. During the initial epochs, the convolutional layers learned visual features such as edges and shapes. As training continues, the improvement rate gradually stabilizes, suggesting convergence.

Simultaneously, the training loss curve steadily decreased across all epochs, indicating successful minimization of the cross-entropy loss function using the Adam optimizer. The smooth decline in loss suggests stable gradient updates and effective learning.

The combination of increasing accuracy and decreasing loss demonstrates that the CNN effectively learned image representations from the Fashion-MNISt dataset. No major fluctuations were observed during training, indicating stable optimization and appropriate hyperparameters selection.

The final CNN training accuracy exceeded 96%, while the testing accuracy remained strong at approximately 91.31%, suggesting good generalization to unseen data.

In contrast, the RF classifier achieved perfect training accuracy but lower testing performance suggesting overfitting. This highlights the advantages of CNNs in preserving spatial information and learning generalized image features directly from two-dimensional pixel structures.

4.3 Confusion Matrices

Confusion matrices were generated for both the CNN and the RF classifier to evaluate category-level classification performance on the Fashion-MNIST dataset. These matrices visualize correct and incorrect predictions across all ten fashion categories.

The results show that both classifiers performed well for visually distinctive categories such as Trouser, Sandal, Sneaker, Bag, and Ankle boot, which have clear geometric patterns.

However, both classifiers struggled with upper-body clothing categories, especially Shirt, Pullover, Coat, and T-shirt/top. These classes contain overlapping shapes and low-resolution greyscale patterns, leading to frequent misclassification. Shirt images were often misclassified as T-shirt/top or Pullover, while the Coat images were occasionally confused with Pullover or Shirt.

The RF classifier showed more confusion between similar categories because flattening images into one-dimensional pixel vectors removes spatial relationships between neighboring pixels, causing the model to rely on overall pixel intensity rather than local structure.

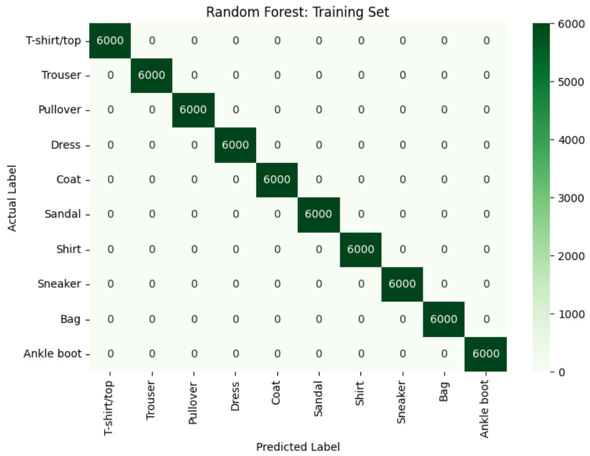
In comparison, the CNN achieved better class separation with fewer misclassifications, particularly for Coat and Pullover classes. This improvement is due to convolutional layers preserving spatial structures and learning local visual patterns such as edges and shapes.

Overall, the confusion matrix analysis supports the overall accuracy findings, showcasing that the CNN achieved stronger generalization performance and more balanced classification behavior across all categories compared to the RF classifier.

Figures 4 to 7 present the confusion matrices for the training and test set of both classification models:

Fig.3

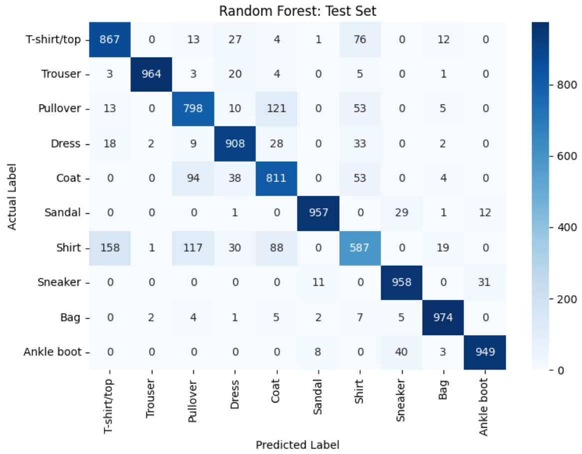
Random Forest Training Confusion Matrix



Note: From "CV_MNIST_FASHION.ipynb" file, section 4. Source: finding of author

Fig.4

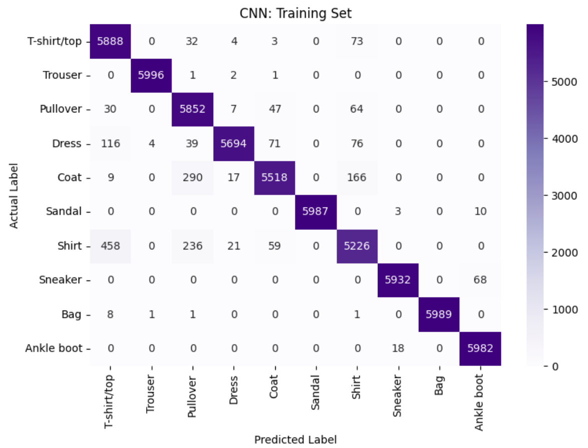
Random Forest Test Confusion Matrix



Note: From "CV_MNIST_FASHION.ipynb" file, section 4. Source: finding of author

Fig.5

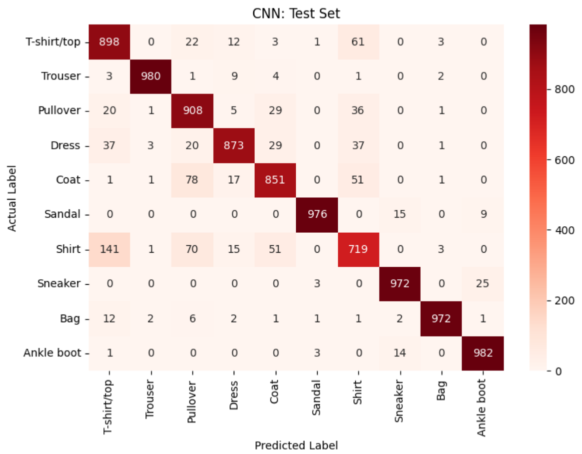
CNN Training Confusion Matrix



Note: From "CV_MNIST_FASHION.ipynb" file, section 4. Source: finding of author

Fig.6

CNN Test Confusion Matrix



Note: From "CV_MNIST_FASHION.ipynb" file, section 4. Source: finding of author

4.4 Precision, Recall, F1-Score, and Accuracy Analysis

To provide a more detailed evaluation of classification performance, precision, recall, and F1-score alongside the overall accuracy. While accuracy provides a general measure of performance, these metrics offer a deeper insight into the category-level prediction quality.

Precision measures the proportion of correctly predicted samples within a class, while recall measures the proportion of correctly identified actual samples. The F1-score represents the harmonic mean of both metrics.

Both classifiers achieved strong results for visually distinctive categories such as Trouser, Sandal, Sneaker, Bag, and Ankle boot, with several CNN F1-scores exceeding 0.96. These categories contain clearer geometric structures and more distinguishable visual patterns

Lower scores were observed for Shirt, Pullover, Coat, and T-shirt/top due to overlapping greyscale features and similar visual structures. The RF classifier showed the greatest difficulty with the Shirt category, achieving substantially lower recall and F1-Score values than the CNN.

Overall, the CNN consistently outperformed the RF classifier across the most difficult categories. In particular, the CNN achieved improved recall for Coat and Pullover, indicating stronger spatial feature extraction capability and improved ability to separate visually similar garments.

The overall accuracy results further support the superiority of the CNN mode. The CNN achieved a testing accuracy of 91.31%, whereas the RF classifier achieved 87.73%, indicating that the CNN provided a more balanced and generalized classification performance.

Tables 2 and 3, showcase the precision, recall, and F1-score values obtained by both classifiers on the Fashion-MNIST dataset:

Table 2: ***RF Classification Repost (Test Set)***

	Precision	**Recall**	**F1-Score**
T-shirt/top	0.82	0.87	0.84
Trouser	0.99	0.96	0.98
Pullover	0.77	0.80	0.78
Dress	0.88	0.91	0.89
Coat	0.76	0.81	0.79
Sandal	0.98	0.96	0.97
Shirt	0.72	0.59	0.65
Sneaker	0.93	0.96	0.94
Bag	0.95	0.97	0.96
Ankle boot	0.96	0.95	0.95

Note: From "CV_MNIST_FASHION.ipynb" file, section 4. Source: finding of author

RF Overall Test Accuracy : 87.73% ≈ 88%

Table 3: ***CNN Classification Repost (Test Set)***

	Precision	**Recall**	**F1-Score**
T-shirt/top	0.81	0.90	0.85
Trouser	0.99	0.98	0.99
Pullover	0.82	0.91	0.86
Dress	0.94	0.87	0.90
Coat	0.88	0.85	0.86
Sandal	0.99	0.98	0.98
Shirt	0.79	0.72	0.75
Sneaker	0.97	0.97	0.97
Bag	0.99	0.97	0.98

Ankle boot	0.97	0.98	0.97

Note: From "CV_MNIST_FASHION.ipynb" file, section 4. Source: finding of author

CNN Overall Test Accuracy : 91.71% ≈ 92%

4.5 Worst Classes Analysis

The most difficult categories to classify were Shirt, T-shirt/top, Pullover, And Coat. Both classifiers produced lower precision, recall and F1-sore values for these categories compared to visually distinctive classes such as Trouser, Bag, Sneaker, and Ankle boot. The confusion matrices also showed frequent misclassification between these upper-body clothing categories

Among all categories, Shirt produced the weakest performance for both classifiers. The RF model achieved a recall value of only 0.59, while the CNN improved recall to approximately 0.72. This suggests strong visual overlap between Shirt, T-shirt/top, Pullover, and Coat classes.

One major reason for the reduced performance is the low-resolution structure of the Fashion-MNIST dataset. Since each image contains only 28x28 greyscale pixels, important visual details such as the fabric textures, collars, and stitching are not clearly represented. As a result, the models rely primarily on general outlines and silhouette structures.

The RF classifier experienced greater difficulty because flattening images into one-dimensional vectors removed spatial relationships between the neighboring pixels, limiting its ability to learn local visual structures. In contrast, the CNN preserved spatial information through convolutional operations and extracted hierarchical features such as edges and shapes, allowing better separation between visually similar garments.

Overall, the worst-class analysis demonstrates the importance of spatial feature extraction in fashion image classification and further supports the superiority of CNN architectures when distinguishing between visually similar fashion products.

4.6 Overfitting Discussion

The experimental results revealed clear differences in overfitting behaviors between the CNN and the RF classifier. Overfitting occurs when a model memorizes the training data instead of learning generalized patterns applicable to unseen data. This is identified when the training accuracy becomes significantly higher than the testing accuracy.

The RF classifier demonstrated substantial overfitting during training. Although the model achieved 100% training accuracy, the testing accuracy decreased to 87.73%, producing a noticeable performance gap. This suggests that the model relied on memorized patterns rather than learning generalized image representations.

This behavior is related to decision tree-based learning, where highly specific decision boundaries are created to minimize training error. Additionally, flattening Fashion-MNIST image into one-dimensional vectors removed important spatial relationships between neighboring pixels, reducing the model's ability to identify generalized structural patterns.

In comparison, the CNN demonstrated stronger generalization capability. The CNN achieved 96.77% training accuracy and 91.31% testing accuracy, resulting in a smaller performance gap between training and testing datasets. This indicates that the CNN learned meaningful hierarchical features rather than simply memorizing training samples.

The convolutional layers extracted local visual features, while max-pooling reduced dimensionality and improved robustness to small image variation. The data was then flattened and passed to a dense output layer with a softmax activation to perform the final classification.The training curves also demonstrated stable learning behavior through optimization, with steadily increasing accuracy and decreasing loss values throughout training.

Overall, the overfitting analysis demonstrates that the CNN provided more reliable and generalized classification performance than the RF classifier, supporting the suitability of the CNN architectures for complex image recognition tasks involving spatial relationships and visually similar objects.

5. Discussion

5.1 CNN vs RF

The experimental results demonstrate a clear performance difference between the CNN and the RF classifier for the Fashion-MNIST image classification. Although both models achieved relatively strong results, the CNN consistently outperformed the RF in terms of testing accuracy, class-level prediction reliability, and generalization capability.

The CNN achieved a testing accuracy of 91.31%, whereas the RF classifier achieved 87.73%. This difference is mainly due to the CNN's ability to preserve spatial relationships and automatically learn hierarchical image features such as edges, textures, and shape. These features are important for distinguishing visually similar categories such as Shirt, Pullover, etc.

In contrast, the RF classifier processed flattened one-dimensional image vectors, removing important spatial information and limited its ability to recognize local visual structures. As a result, RF produced more misclassifications with overlapping clothing features.

The result also revealed significant differences in generalization performance. RF achieved a perfect training accuracy but lower testing accuracy, indicating substantial overfitting. In contrast, the CNN maintained a smaller gap between training and testing accuracy, suggesting stronger generalization capability.

Despite its lower predictive performance, RF showed several advantages. The model was simpler to implement, required fewer computational resources, and completed training significantly faster than the CNN. RF models are generally easier to interpret compared to deep neural networks.

Overall, the analysis demonstrated that CNNs are more reliable for image classification tasks involving visually similar categories due to their ability to preserve spatial information and learn complex visual features.

5.2 Computational Performance

Computational performance was evaluated using training time and inference time.

The RF classifier demonstrated significantly faster computational performance than the CNN. RF completed training in 23.3 seconds, while the CNN required approximately 302.1 seconds due to convolutional operations, backpropagation, and iterative optimization across multiple epochs.

Inference time analysis also produced similar observations. RF generated predictions in approximately 0.20 seconds, compared to the approximately 1.43 seconds for CNN. Although both inference times remained relatively low, RF demonstrated superior computational efficiency.

Despite its increased computational costs, the CNN achieved stronger predictive accuracy and better generalization capability. Therefore, the higher training cost may be justified for applications where the classification accuracy and reliability are prioritized over computational efficiency.

Overall, the computational analysis highlights the trade-off between predictive performance and computational efficiency. RF offers faster training and lower resource usage, whereas CNN delivers superior image classification performance at the expense of the increased computational complexity.

5.3 Production Recommendations

Based on the experimental findings, the CNN is recommended for practical fashion image classification systems. Although RF demonstrated faster computational performance and lower training complexity, the CNN achieved stronger predictive accuracy and better generalization capability.

The CNN achieved fewer misclassifications for visually similar categories such as Shirt, T-shirt/top, Coat, and Pullover. This is particularly important in real-world retail applications, where incorrect product categorization may negatively affect inventory management, recommendation systems, and customer search experiences.

One major advantage of CNN architecture is its ability to automatically learn hierarchical spatial features directly from the image data through convolutional operation. This allows CNNs to perform better on image recognition tasks involving overlapping visual patterns.

Although the CNN required substantially longer training time and higher computational resources, modern production environments commonly use Graphics Processing Units (GPUs) to accelerate DL workloads. Therefore, the increased computational cost can be justified by its superior predictive performance and improved reliability.

RF classifiers may still be suitable for smaller-scale applications where the computational resources are limited or rapid baseline implementation is required.

Nevertheless, the findings support the use of CNN architectures for practical fashion image classification systems within retail and e-commerce environments.

5.4 Reflection of Methods

The methodology used in this project enabled a comprehensive comparison between a DL approach and traditional ML approach for fashion image classification. Using the Fashion-MNIST dataset allowed benchmarking under balanced and visually challenging classification conditions.

A strength of the methodology was the use of consistent preprocessing, training, and evaluation procedures for both models. Multiple evaluation metrics, including accuracy, precision, recall, F1-Score, confusion matrices, and computational performance, provided a comprehensive assessment beyond overall accuracy alone.

However, several limitations remain. Fashion-MNIST dataset contains low-resolution greyscale images that do not fully represent the complexity of real-world fashion product images, which often contain varying lighting conditions, backgrounds, image resolutions, and color information that may significantly influence classification performance.

Additionally, only one CNN architecture and one RF configuration were evaluated. Alternative CNN architectures, transfer learning methods, or hyperparameter optimization may further improve classification performance.

Another limitation was the absence of data augmentation techniques, such as rotation, flipping, scaling, and translation which could improve CNN robustness and reduce overfitting.

Despite these limitations, the methodology successfully fulfilled the objectives of the study by providing a systematic comparison between the DL and traditional ML approaches for the fashion image classification. The findings demonstrate the importance of spatial feature extraction and highlight the effectiveness of convolutional architectures for complex image recognition tasks.

6. Conclusion

This project evaluated the performance of a Convolutional Neural Network (CNN) and a Random Forest (RF) classifier for fashion product classification using the Fashion-MNISt dataset. The study evaluated a deep learning (DL) approach with a traditional machine learning (ML) approach in terms of classification accuracy, generalization capability, computational performance, and category-level prediction behavior.

The results showcased that the CNN achieved superior overall performance, reaching a testing accuracy of 91.31%, while the RF classifier achieved 87.73%. The CNN also produced more balanced precision, recall, and F1-score value across most fashion categories, particularly for visually similar clothing items such as Coats, Shirts, T-shirt/top, and Pullover.

The analysis highlighted important differences in feature learning between the two models. The CNN preserved spatial relationships and extracted hierarchical visual features through convolutional operations, enabling stronger generalization to unseen data. In contrast, the RF classifier processed flattened one-dimensional image vectors, removing important spatial information and increasing misclassification rates between visually similar categories.

The RF classifier also identified clear overfitting behavior, achieving perfect training accuracy but noticeably lower testing accuracy. The CNN maintained a smaller gap between training and testing accuracy score, demonstrating stronger robustness and more reliable predictive behavior.

From a computational perspective, the RF classifier required significantly less training time and computational resources. However, the improved predictive performance and generalization capability of the CNN outweighed its higher computational cos, making it the preferred solution for the fashion image classification system.

Overall, the findings demonstrate the effectiveness of CNN architectures for image recognition tasks involving visually similar objections and highlight the importance of spatial feature extraction in computer vision application. Further research may improve performance by exploring deeper CNN architectures, transfer learning techniques, image augmentation strategies, and larger real-world fashion datasets containing color images and more diverse visual conditions.

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8. Appendix: The code

Github code link : https://github.com/maryamrad99/Project-Computer-Vision-Fashion-MNIST